Performance Testing Report

Computer Systems and Network Engineering — Final Year Research Project

Project Title:

OceanSense: An IoT-Based Hybrid System for Real-Time Detection and Tracking of Marine Debris

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1.Introduction and Objective

The OceanSense project develops an IoT-based swarm of Autonomous Surface Vehicles (ASVs) for real-time detection and tracking of marine debris in dynamic ocean environments, using low-power LoRa communication for inter-node coordination. The environment-aware adaptive swarm coordination component is central to enabling decentralised decision-making and efficient area coverage across the designated search region.

Performance testing is essential to evaluate how effectively the swarm coordination algorithm operates under realistic environmental and communication constraints. Ocean conditions — including varying wave heights, currents, and uneven debris distribution — introduce significant challenges to maintaining stable formations and achieving efficient coverage. The limited bandwidth and intermittent nature of LoRa communication further restrict coordination between nodes.

This report evaluates the adaptive swarm coordination mechanism with respect to its ability to maintain coverage efficiency, formation stability, and communication effectiveness. It analyses how the swarm adapts to environmental changes and how communication constraints impact coordination behaviour. The objective is to assess whether the proposed environment-aware coordination strategy enhances system efficiency and robustness under varying conditions, while operating within the limitations of decentralised control and low-bandwidth communication.

2.Performance Testing Environment Setup

2.1 Hardware Configuration

Testing was conducted on a simulation-based platform: Intel Core i5 11th Gen, 16 GB RAM, 256 GB NVMe SSD, running Windows 11.

2.2 Software Configuration

Webots Simulator modelled the marine environment including ASV dynamics, wave conditions, and debris distribution. **Robot Operating System (ROS)** implemented decentralised control logic with each ASV as an independent ROS node. Algorithms and data processing were developed in **Python** (scripting and analysis) and **C++** (performance-critical control loops).

2.3 Network Conditions and Communication Model

A custom LoRa-emulation layer introduced controlled message delays, limited transmission frequency, and simulated packet loss to replicate real-world constrained communication conditions within the ROS environment.

3.Tools and Frameworks Used

Webots served as the primary simulation platform, providing a physics engine capable of modelling dynamic marine conditions — including waves and currents — critical for realistic swarm evaluation. **ROS** underpinned the decentralised coordination logic, with each ASV operating as an independent ROS node, enabling distributed decision-making consistent with the system design. **Python** was primarily used for rapid development, scripting, and data analysis, while **C++** was applied where performance efficiency was required, particularly in control loops and simulation integration. Custom scripts captured node positions, communication events, and coverage and stability metrics over time, with post-processing tools computing final metrics and generating the performance visualisations used in Section 6.

5.Testing Methodology and Execution

4.1 Test Scenarios

Five scenarios were defined to cover the range of environmental and communication conditions affecting swarm behaviour:

- **S1 – Baseline (Non-Adaptive):** Fixed formation with no environmental awareness; serves as the reference benchmark.
- **S2 – Adaptive, Calm Conditions:** Stable environment, low waves, uniform debris; establishes ideal adaptive performance.
- **S3 – High Wave Conditions:** Increased wave intensity; swarm adapts inter-node spacing to preserve formation stability.
- **S4 – Ocean Current Influence:** Directional current applied; swarm reorients to maintain effective coverage.
- **S5 – Limited Communication (LoRa Constraints):** Packet loss and message delays; tests robustness of decentralised coordination.

4.2 Execution Procedure

The simulation environment was initialised in Webots, the swarm deployed with predefined parameters, environmental conditions configured per scenario, and the coordination algorithm activated. Custom logging tools recorded performance data throughout the run. Each scenario was repeated to ensure reproducibility.

5.Metrics Measured

Area Coverage Efficiency measures the percentage of the total designated region visited or monitored by the swarm over time, reflecting how effectively the swarm explores its environment. **Formation Stability** is quantified by the variance in inter-node distances over time — lower variance indicates the swarm is maintaining its intended structure under disturbances. **Communication Success Rate** is the ratio of successfully received messages to total messages transmitted, capturing the reliability of LoRa-constrained inter-node communication.

6.Results and Analysis

6.1 Results

Scenario	Coverage (%) ↑	Variance ↓	Comm. (%) ↑	Performance Insight
Baseline	67	3.0	93	Poor adaptability; instability increases over time
Calm Adaptive	87	0.6	95	Optimal performance under stable conditions
High Wave	84	0.8	89	Slight instability from environmental disturbance
Current	92	0.5	91	Best coverage via directional alignment
Low Comm	78	2.7	72	Performance degraded by communication constraints

Table 1: Performance metrics across all test scenarios.

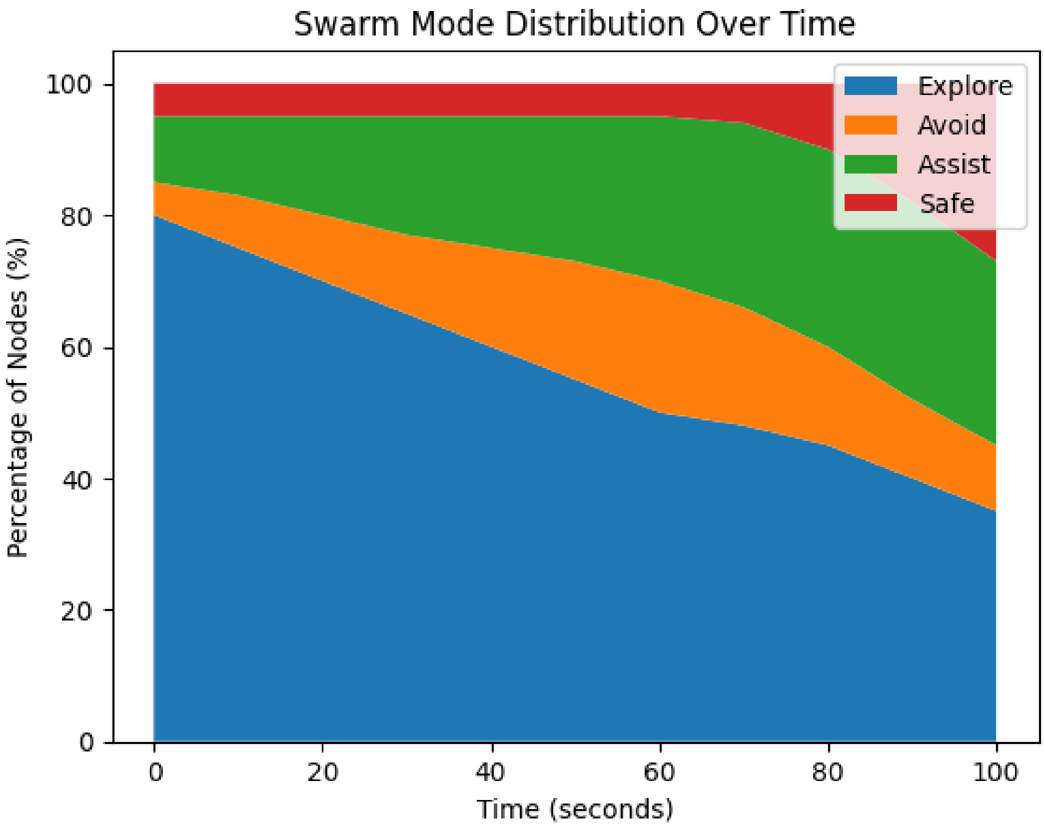

Figure 1: Swarm mode distribution over time illustrating transitions between Explore, Avoid, Assist, and Safe states under varying environmental conditions.

6.2 Analysis of Area Coverage

The adaptive algorithm outperformed the baseline across all challenging scenarios. Under calm conditions, both approaches achieved comparable coverage, though the adaptive method converged marginally faster via dynamic node distribution. In high-wave conditions, adaptive inter-node spacing reduced overlap and improved distribution (84% vs 67% baseline). The current scenario achieved the highest coverage (92%) by aligning swarm movement with the directional current.

6.3 Analysis of Formation Stability

Stability improved markedly with the adaptive approach. Distance variance dropped from 3.0 (baseline) to 0.8 under high waves, with dynamic spacing absorbing environmental disturbances. Under communication constraints, variance rose to 2.7 as delayed updates degraded coordination; however, local sensing allowed nodes to maintain acceptable formation behaviour.

6.4 Impact of Communication Constraints

Communication success dropped to 72% under LoRa constraints, causing occasional coordination inconsistencies. Delayed messages produced outdated neighbour information, slightly reducing stability. Despite this, the decentralised design allowed nodes to continue operating using local data, minimising overall performance degradation.

7.Performance Optimization and Recommendations

7.1 Optimization Techniques

Four optimisations were identified: (i) **Adaptive inter-node spacing** based on wave intensity improves stability and reduces collision risk; (ii) **event-driven communication** triggered by significant events (debris detection, formation deviation) reduces overhead under low bandwidth; (iii) **reduced communication dependency** through stronger local sensing improves robustness under packet loss; and (iv) **adaptive orientation based on currents** sustains coverage efficiency during directional disturbances.

7.2 Recommendations for Future Improvements

Future work should evaluate scalability with larger node counts, incorporate more realistic LoRa models (duty-cycle enforcement, interference), and integrate energy-aware scheduling with coordination logic.

8.Challenges and Solutions

LoRa Simulation: Standard ROS communication assumes high-bandwidth links, which does not reflect LoRa behaviour. A custom layer with controlled delays, limited transmission rates, and packet loss simulation was implemented to address this.

Formation Stability Under Disturbances: Waves and currents caused node position fluctuations. Adaptive spacing mechanisms were incorporated into the coordination algorithm to dynamically adjust relative node positions based on environmental conditions.

Decentralised Synchronisation: Communication delays caused inconsistent neighbour information. The algorithm was redesigned to prioritise local sensing and periodic updates, reducing dependency on synchronised global state.

Performance Data Collection: Continuous multi-node monitoring was not natively supported by the simulation tools. Custom logging scripts were developed to record positions and communication events, with post-processing pipelines computing final metrics.

9.Ethical and Compliance Considerations

All testing was conducted entirely within a simulated environment using synthetic data generated in Webots. No real-world users, personal data, or sensitive information were involved, and no privacy or data protection concerns arise from this work. Webots and ROS are open-source platforms; their licences have been observed throughout the project. All tools, frameworks, and references have been acknowledged in accordance with academic standards.

10.References

1. Webots Robot Simulator. Open-source mobile robot simulation software. https://cyberbotics.com

2. Robot Operating System (ROS). Open-source robotics middleware. https://www.ros.org

3. LoRa Alliance. LoRaWAN Specification. https://lora-alliance.org/resource_hub/lorawan-specificat

4. Brambilla, M., Ferrante, E., Birattari, M., & Dorigo, M. (2013). Swarm robotics: A review from the swarm engineering perspective. *Swarm Intelligence*, 7(1), 1–41.

5. Dorigo, M., Theraulaz, G., & Trianni, V. (2021). Swarm robotics: Past, present, and future. *Proceedings of the IEEE*, 109(7), 1152–1165.

6. Queralta, J. P., et al. (2020). Collaborative multi-robot search and rescue: Planning, coordination, perception, and active vision. *IEEE Access*, 8, 191617–191643.

7. Sousa, J., et al. (2016). LSTS: A toolchain for operating multiple vehicle types in networked environments. *IFAC-PapersOnLine*, 49(23), 408–413.